

Thermax Neo demonstrated the true potential of VSA technology in biogas purification and upgradation



Industries

Biogas/Bio CNG/Renewable Natural Gas (RNG)

Challenge

In India, VSA technology has faced skepticism due to adsorbent degradation, methane loss, and scalability issues, limiting its acceptance in the biogas industry.

Solution

At one of India's largest bio-gas projects, Thermax Neo installed a 2 TPD in-house VSA-based biogas upgradation unit that processes cow dung, press mud, and paddy straw into purified biogas.

Company Profile

India's largest renewable energy solutions provider

Key Achievements



High Methane Purity

The unit has consistently delivered enriched methane with over 96% purity.



Low Methane Slippage

Methane loss was kept below 3%, meeting industry benchmarks.



Product Gas Applications

The upgraded biomethane is compressed and supplied to CNG stations for use as automotive fuel.

Thus, vacuum swing adsorption (VSA) technology offers an efficient solution for biogas upgrading, balancing operational simplicity, high methane recovery, and cost-effectiveness. Additionally, the successful operation serves as a model for other small and medium-scale biogas facilities in India, showcasing the adaptability and efficiency of VSA technology.

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